

Luis 'Mateo' Gutierrez



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PAPERS

Gutierrez, M., Tumuklu, O. (2025). *Characterization of the Compressible Kelvin-Helmholtz Instability*. AIAA SciTech Conference.
Gutierrez, M., (2025). *Viscous and Nonequilibrium Effects on Kelvin-Helmholtz Instabilities*. RPI Master's Thesis.

WORK EXPERIENCE

CURRENT, FROM NOV 2025 (PT)

Design Build Fly *Wing Tip Simulation*

Performing CFD analysis using OpenFOAM on a wing configuration including wing tip flow characterization and symmetric half-body simulations to support iterative design decisions.

CURRENT, FROM JAN 2025 (FT)

HAVA Lab *Graduate Researcher*

Facilitated the porting of a C++ library, Hy2Foam, to an updated version of OpenFOAM. Implemented various Riemann solvers into Hy2Foam including HLL, HLLC, AUSM+ and AUSM+UP.

JAN 2024 – MAY 2024 (FT)

Senior Capstone *Process Automation for Commons Dining Hall Dishroom*

Implemented a system that automates operations in one of RPI's dining halls, reducing labor-intensive tasks performed by employees and enhancing dish throughput efficiency. Led the design of a pre-cleaning station with water jets, utilizing Advsim 3D for design and iteration.

MAR 2023 – AUG 2023 (FT)

3LP Engineering *Structural Engineering Intern*

Coordinated with Project Managers, Architects, and other engineering disciplines to accomplish over twenty-five projects valued over \$25,000 in a timely and efficient manner. Managed a project portfolio that included structural design and analysis for a variety of residential and commercial projects of various building materials. Utilized AutoCAD to draft structural residential and commercial structural plans and details.

FEB 2023 – MAY 2023 (PT)

Thorben Consulting LLC *Cloud Practitioner*

Collaborated with a team of four working on an early chatbot prototype that utilizes both Machine Learning and AWS Services. Successfully completed the following courses: AWS Cloud Quest: Cloud Practitioner and AWS Cloud Practitioner Essentials.

GRADUATE RESEARCH

“Nonlinear Modal Stability Formulation for Hypersonic Nonequilibrium Flows”

To explore hypersonic instabilities arising from complex shock wave boundary layer interactions (SWBLI) with varying model geometries, particularly focusing on double wedges and cones. This investigation will utilize open-source computational fluid dynamics software tools such as Gmsh and OpenFOAM.

“Examining Kelvin Helmholtz Instabilities utilizing Riemann Solvers and WENOEXT in OpenFOAM”

Investigating the implementation of the Weighted Essentially Non-Oscillatory (WENO) scheme as well as Riemann solvers in the OpenFOAM solvers rhoCentralFoam and hy2Foam to validate the effectiveness of WENO and Riemann numerical methods in resolving instabilities for a Kelvin-Helmholtz validation case.

EDUCATION

2020-2024 **Bachelor of Science**
GPA: 3.32
Mechanical Engineering
Rensselaer Polytechnic Institute

2024-2025 **Master's of Science**
GPA: 3.52
Aeronautical Engineering
Rensselaer Polytechnic Institute

HONORS & AWARDS

2019 **Eagle Scout**
Boy Scouts of America

2020-CURRENT **RPI Leadership Scholarships**
Rensselaer Polytechnic Institute

2025 **Pi Lambda Phi Scholarship Key**
Rensselaer Polytechnic Institute

COMPUTER SKILLS

COMPUTATIONAL	OpenFOAM, hy2Foam, Python Matlab, ParaView, GMSH
CAD	NX, Solidworks
MISC	Linux OS, Windows OS, L ^A T _E X
LEARNING SKILLS	OpenFoam Development (C++) FEniCSx

RELEVANT COURSEWORK

CLASSES	Propulsion Systems, Incompressible Flow Compressible Flow, Graduate Fluid Mechanics Finite Elements Methods, Computational Fluid Dynamics Numerical Design Optimization
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